

Working of the universal TM U .

U takes as input an encoding of a TM M and a string x and simulates M on x .

- halts and accepts if M halts and accepts x
- halts and rejects if M halts and rejects x .
- loops if M loops on x .

U simulates M step by step.

Question. Can we do better than blind simulation?

Eg. if M halts on x then simulate M on x
if M does not halt on x then terminate the simulation and reject.

That is: Build U' that takes as input $M \# x$ and

- halts and accepts if M halts and accepts x
- halts and rejects if M halts and rejects x .
- halts and rejects if M loops on x .

Thus $L(U') = L(U) = \underbrace{\{M \# x \mid x \in L(M)\}}_{\text{recursive?}}$

$$HP = \{M \# x \mid M \text{ halts on } x\}$$

Question. Is HP recursive?

Does there exist a total TM M s.t. $L(M) = HP$

Answer. No

Question. Is HP recursively enumerable (r.e.)?

Does there exist a TM M s.t. $L(M) = HP$?

Answer. Yes.

$$MP = \{M \# x \mid x \in L(M)\}$$

Question. Is MP recursively enumerable (r.e.)?

Does there exist a TM M s.t. $L(M) = MP$?

$$HP = \{m \# x \mid m \text{ halts on } x\}$$

$$MP = \{m \# x \mid x \in L(m)\}$$

Question. Is HP recursive?

Does there exist a total TM M s.t. $L(M) = HP$

Answer. No

Question. Is MP recursive?

Does there exist a total TM M s.t. $L(M) = MP$

Answer. No.

Theorem. HP is not recursive.

For $x \in \{0,1\}^*$, let M_x denote the TM with input alphabet $\{0,1\}$ whose encoding is x .

	ϵ	0	1	00	01	11	000	001	--
M_ϵ	H	L	H	H	L	L	L	H	--
M_0	L	H	L	L	H	L	H	H	
M_1	H	L	H	H	L	H	L	H	
M_{00}	L	L	L	H	L	H	H	H	
M_{01}	H	H	L	H	H	L	L	L	
M_{11}	L	L	H	H	H	L	H	L	
M_{000}	L	H	L	H	L	H	H	L	
M_{001}	H	L	H	H	L	H	H	H	
$\vdots$									

$-x^{\text{th}}$ row describes for input y if M_x halts on y .

	ϵ	0	1	00	01	11	000	001	--
M_ϵ	H	L	H	H	L	L	L	H	--
M_0	L	H	L	L	H	L	H	H	
M_1	H	L	H	H	L	H	L	H	
M_{00}	L	L	L	H	L	H	H	H	
M_{01}	H	H	L	H	H	L	L	L	
M_{11}	L	L	H	H	H	L	H	L	
M_{000}	L	H	L	H	L	H	H	L	
M_{001}	H	L	H	H	L	H	H	H	

Suppose $\exists$ a total TM K such that $L(K) = HP$

For any x and y , K can determine the entry in the $(x,y)^{th}$ cell in the above table.

On input $M\#x$,

- $\bullet$ K halts and accepts if M halts on x
- $\bullet$ K halts and rejects if M loops on x

Consider a TM N that on input $x \in \{0,1\}^*$ does the following:

- 1) Constructs M_x from x and writes $M_x\#x$ on its tape
- 2) Runs K on input $M_x\#x$

	ϵ	0	1	00	01	11	000	001	--
M_ϵ	H	L	H	H	L	L	L	H	--
M_0	L	H	L	L	H	L	H	H	
M_1	H	L	H	H	L	H	L	H	
M_{00}	L	L	L	H	L	H	H	H	
M_{01}	H	H	L	H	H	L	L	L	
M_{11}	L	L	H	H	H	L	H	L	
M_{000}	L	H	L	H	L	H	H	L	
M_{001}	H	L	H	H	L	H	H	H	

Suppose $\exists$ a total TM K such that $L(K) = HP$

For any x and y , K can determine the entry in the $(x,y)^{th}$ cell in the above table.

On input $M\#x$,

- K halts and accepts if M halts on x
- K halts and rejects if M loops on x

Consider a TM N that on input $x \in \{0,1\}^*$ does the following:

- 1) Constructs M_x from x and writes $M_x\#x$ on its tape
- 2) Runs K on input $M_x\#x$, accepting if K rejects and going into a trivial loop if K accepts.

For any $x \in \{0,1\}^*$, N halts on x iff K rejects $M_x\#x$
iff M_x loops on x .

That is, N is different from every M_x on at least one string - the string x .
This gives a contradiction.